

Tuesday, April 28

Exam:

50 79/79.5 100
 ←————→
 15

↑
Hypothetical

Grades: All B- or above

(details on Discord and Canvas)

Plan

Proposals due

→ I'll message w/ feedback

Fast JL Transform today

Explainable AI next

Talks

Ahmet 4-5pm Wednesday @ Davidson

Feyza 4:15-5:15pm Friday @ Estelle 1021

Regression Review

$$A \in \mathbb{R}^{n \times d}, b \in \mathbb{R}^n$$

$$x^* = \operatorname{argmin}_{x \in \mathbb{R}^d} \|Ax - b\|_2^2 \quad \text{takes } O(nd^2) \text{ time to solve}$$

Let $m \ll n$ where $m \approx O(d)$, $\Pi \in \mathbb{R}^{m \times n}$ JL transform

$$\tilde{x} = \operatorname{argmin}_{x \in \mathbb{R}^d} \|\Pi Ax - \Pi b\|_2^2 \quad \text{takes } O(md^2) \text{ to solve}$$

Today: Build Π so that

IF $\nearrow$ we already have ΠA

Time to compute: $O(mnd) \approx O(nd^2)$

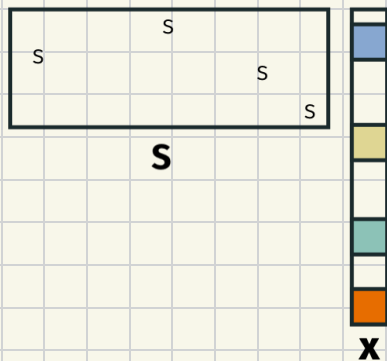
1. $(1-\epsilon)\|x\|_2 \leq \|\Pi x\|_2 \leq (1+\epsilon)\|x\|_2$ whp

2. Compute Πx in $O(n \log n)$ time

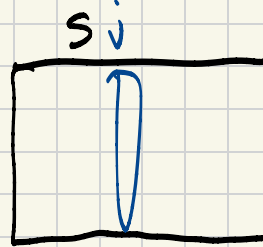
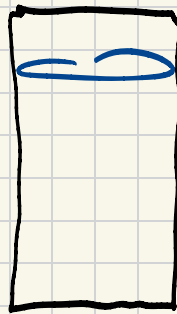
Sampling Matrix

$$S \in \mathbb{R}^{m \times n}$$

$$\text{Time: } O(m)$$



i

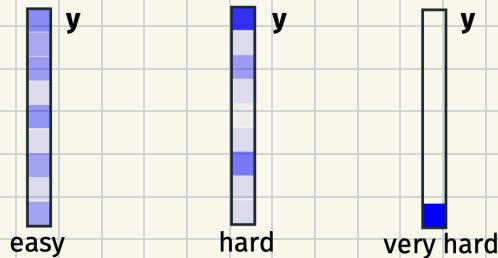


Every row contains a single value $s = \sqrt{\frac{n}{m}}$ in a random column

$$\begin{aligned} \mathbb{E} \|Sx\|_2^2 &= \mathbb{E} x^T S^T S x \\ &= x^T \mathbb{E} [S^T S] x \\ &= x^T x \\ &= \|x\|_2^2 \end{aligned}$$

$$\begin{aligned} \mathbb{E} [(S^T S)_{ij}] &= \mathbb{E} \sum_{k=1}^m [S^T]_{i,k} [S]_{k,j} \\ &= \mathbb{E} \sum_{k=1}^m \mathbb{1}[i \text{ in row } k] \mathbb{1}[j \text{ in row } k] \frac{n}{m} \\ &= \sum_{k=1}^m \Pr(i \text{ in row } k, j \text{ in row } k) \frac{n}{m} \\ &= \begin{cases} 1 & \text{if } i=j \\ 0 & \text{else} \end{cases} \end{aligned}$$

Concentration



Claim: If $x_i^2 \leq \frac{c}{n} \|x\|_2^2$ for all i ,
 $m = O(c \log(1/\delta) / \epsilon^2)$ preserve ℓ_2 -norm whp.

Challenge: Guarantee $x_i^2 \leq \frac{c}{n} \|x\|_2^2$ whp.

Build M so that

1. $\|Mx\|_2^2 = \|x\|_2^2$

2. $[Mx]_i^2 \leq \frac{c}{n} \|x\|_2^2$

3. Compute Mx in $O(n \log n)$ time

$$\Leftrightarrow M^T M = I \Leftrightarrow M^{-1} = M^T$$

so $\exists y$ s.t. $My = e_i \Leftrightarrow y = M^{-1}e_i$

very hard
input so make
 M random

$M = HD$ where $D \in \mathbb{R}^{n \times n}$ is diagonal with ± 1
 $H \in \mathbb{R}^{n \times n}$ is Hadamard

Assume n is a power of 2.

$$H_0 = [1]$$

$$H_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} H_1 & H_1 \\ H_1 & -H_1 \end{bmatrix}$$

$\vdots$

$$H_k = \frac{1}{\sqrt{2}} \begin{bmatrix} H_k & H_k \\ H_k & -H_k \end{bmatrix}$$

Each entry is $\pm \frac{1}{\sqrt{2}^k} = \pm \frac{1}{\sqrt{n}}$

Property ①: $H_k^T H_k = I \Leftrightarrow \|H_k\|_2 = \|x\|_2$

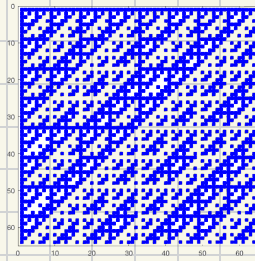
Hint: Use induction

Property ②: Computing $H_k x$ is $O(n \log n)$

Hint: Write $x = \begin{bmatrix} x_a \\ x_b \end{bmatrix}$ and use master method

Property (3): HD is a good mixing matrix

H



HD



Uniform ± 1



SHRT Mixing Lemma: Let $z = HDx$. w.p. $1-\delta$

uses randomization
concentration

$$z_i^2 \leq \frac{C \log \frac{n}{\delta}}{n} \|z\|_2^2.$$

$$\Pi x = S H \underbrace{D}_{O(n)} x$$

$$\underbrace{\quad}_{O(n \log n)}$$

$$O(m)$$

$$(1-\epsilon)\|x\|_2 \leq \|\Pi x\|_2 \leq (1+\epsilon)\|x\|_2 \quad \text{w.h.p.}$$